Project Demo Planning

Date	2 July 2026
Project Name	OptiCrop
Maximum Marks	1 Mark

Project Demo Planning:

OptiCrop is an advanced, data-driven software system that optimizes agricultural production by integrating key environmental factors — Nitrogen (N), Phosphorous (P), Potassium (K), soil temperature, humidity, pH, and rainfall — with crop-specific data. It provides intelligent crop recommendations and suitability insights to help farmers maximize yield and resource efficiency, while also supporting researchers and policymakers in developing evidence-based, sustainable agricultural strategies. This document outlines the plan for demonstrating OptiCrop, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Introduce the challenge of inefficient, non-data-driven crop selection and its impact on yield, resource use, and farmer income; present OptiCrop's goal of using N, P, K, soil, and weather data to close this gap.	3	Ketha Mahidar
2	Solution Overview	Explain the OptiCrop architecture: input parameters (N, P, K, temperature, humidity, pH, rainfall), the recommendation engine, and how it generates crop and suitability insights for farmers, researchers, and policymakers.	3	Ketha Mahidar
3	Live Demo – Smart Crop Recommendation	Scenario 1: Enter soil and environmental readings and show the system recommending the most suitable crop for maximum yield and farming efficiency.	4	Mehar Gannamani
4	Live Demo – Crop Suitability & Environmental Assessment	Scenario 2: Input a specific crop with current soil/climate data and demonstrate the suitability, compatibility, and productivity insights returned.	3	Mehar Gannamani
5	Live Demo – Research & Policy Planning View	Scenario 3: Show the analytics/dashboard view used by researchers and policymakers to study crop-environment patterns and support sustainable farming strategies.	3	Mehar Gannamani
6	Q&A Session	Open the floor for questions, clarify assumptions and data sources used, and summarize key takeaways and future scope of OptiCrop.	2	Mehar Gannamani, Ketha Mahidar

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Set context: why smart, data-driven crop recommendations matter for farmers and sustainable agriculture.
2	Solution Overview	Walk through OptiCrop's inputs (N, P, K, temperature, humidity, pH, rainfall) and how the engine turns them into recommendations.
3	Live Feature Demonstration	Run through all three scenarios: crop recommendation, suitability assessment, and the research/policy view.
4	Q&A Session	Address audience questions and close with a brief recap of impact and next steps.